A HYBRID DEMAND FORECASTING MODEL FOR AN EXPORT-DRIVEN NATURAL FIBRE INDUSTRY

PROJECT REPORT

Submitted by

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TCR24MEIM05

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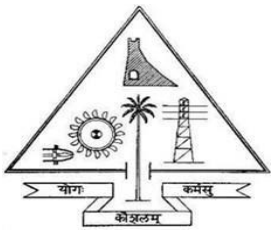
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DECLARATION

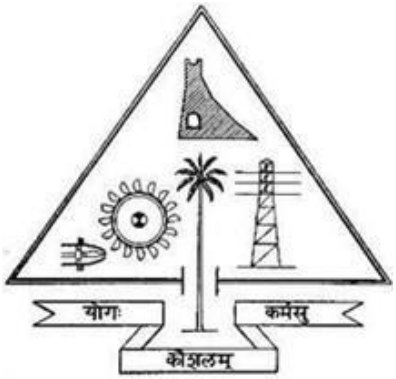
I undersigned, hereby declare that the Project Report titled **A HYBRID DEMAND FORECASTING MODEL FOR AN EXPORT – DRIVEN NATURAL FIBRE INDUSTRY** submitted in complete fulfilment of the requirements for the award of degree of Master of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under the supervision of **Dr.Madhusoodanan M.R**, Professor of mechanical engineering department, GEC Thrissur. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or factor source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other university.

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CERTIFICATE

This is to certify that the Project Report entitled A HYBRID DEMAND FORECASTING MODEL FOR AN EXPORT-DRIVEN NATURAL FIBRE INDUSTRY submitted by **ANTONY JOHN BERCHMANS** Reg.No TCR24MEIM05 to the APJ Abdul Kalam Technological University in complete fulfillment of the requirements for the award of the Degree of Master of Technology in Mechanical Engineering (Industrial Engineering and Management) is a bonafide record of the project work carried out by him/ her under my /our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

Accurate demand forecasting plays a crucial role in improving operational efficiency, production planning, and supply chain management within export-oriented manufacturing industries. The natural fibre sector, characterized by fluctuating market demand, seasonal variations, and nonlinear sales behaviour, requires advanced forecasting techniques capable of capturing complex temporal patterns. This study proposes an intelligent demand forecasting framework that integrates statistical time-series modelling with machine learning approaches to enhance prediction accuracy for natural fibre products.

Historical sales data of three major product categories—Door Mat, Roll Mat, and Geo Textile Mat—were analysed to identify underlying demand characteristics. Comprehensive data preprocessing techniques, including missing value handling, outlier treatment, exploratory data analysis, time-series transformation, and feature scaling, were applied to ensure data quality and modelling reliability. Stationarity analysis using the Augmented Dickey–Fuller (ADF) test revealed the presence of non-stationary behaviour, indicating strong trend and seasonal components within the sales data.

To evaluate forecasting performance, multiple models were implemented, including Autoregressive Integrated Moving Average (ARIMA), Seasonal ARIMA (SARIMA), Extreme Gradient Boosting (XGBoost), Hybrid ARIMA, and Hybrid SARIMA models. Model effectiveness was assessed using standard performance metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). Experimental results demonstrated that standalone statistical models struggled to capture nonlinear demand variations, while machine learning models alone lacked consistent temporal stability.

The proposed hybrid forecasting approach, combining seasonal time-series modelling with machine learning-based residual learning, significantly improved prediction accuracy across all product categories. Among the evaluated models, the Hybrid SARIMA framework achieved the lowest forecasting errors and highest reliability, confirming the effectiveness of hybrid modelling strategies for complex industrial datasets.

The outcomes of this research provide practical insights for demand-driven production planning, inventory optimization, and export management within the natural fibre industry. The study establishes that integrating statistical forecasting with machine learning techniques offers a

robust and scalable solution for real-world industrial demand prediction problems. Furthermore, the proposed framework contributes to the advancement of intelligent forecasting systems applicable to modern data-driven manufacturing environments.

Keywords: Demand Forecasting, Time Series Analysis, ARIMA, SARIMA, XGBoost, Hybrid Models, Machine Learning, Natural Fibre Industry, Sales Prediction.

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CHAPTER 1

INTRODUCTION

The natural fibre industry, particularly the coir sector in Kerala, plays a significant role in sustainable industrial development by utilising coconut husk to produce eco-friendly products such as coir mats, floor coverings, and geotextiles. A substantial portion of coir production is export-oriented, making demand highly sensitive to seasonal trade cycles, international market conditions, and macroeconomic fluctuations. Variations in global demand, exchange rate movements, and shifting export trends introduce complexity and uncertainty into production planning.

Traditional demand forecasting approaches often struggle to capture the non-stationary, seasonal, and non-linear characteristics of export-driven markets. In such environments, reliance on simplistic time-series techniques can lead to inaccurate forecasts, affecting production scheduling and strategic planning. Therefore, there is a need for a more robust and adaptive forecasting framework capable of modelling both linear trend components and complex non-linear demand fluctuations.

With the increasing availability of historical sales data and external market indicators, data-driven forecasting techniques provide an opportunity to enhance predictive reliability. Machine learning algorithms and advanced time-series models can be leveraged to capture intricate temporal dependencies and external demand drivers. This study develops a hybrid demand forecasting framework for an export-driven coir manufacturing environment, integrating statistical modelling, feature engineering, and ensemble learning techniques to improve forecasting accuracy and robustness under volatile market conditions.

1.1 PROBLEM STATEMENT

The coir mat manufacturing industry operates in a highly volatile and export-driven market environment. Demand for coir products is significantly influenced by seasonal cycles, international trade patterns, exchange rate movements, and shifting global consumer preferences. These external dependencies introduce substantial variability and structural complexity into historical sales data.

Traditional forecasting practices in small and medium-scale coir industries often rely on basic statistical techniques that assume stable seasonal behaviour and linear trends. However, export-oriented demand patterns exhibit non-stationarity, nonlinear fluctuations, and sudden structural shifts that conventional models fail to capture effectively. As a result, forecasting errors remain high,

reducing the reliability of production planning and strategic decision-making.

Additionally, limited incorporation of exogenous variables—such as export indices, macroeconomic indicators, and seasonal trade factors—restricts the explanatory power of existing forecasting systems. The absence of advanced feature engineering and hybrid modelling strategies further constrains predictive robustness in dynamic industrial contexts.

Therefore, there is a need to develop a comprehensive, data-driven demand forecasting framework that integrates statistical time-series modelling, machine learning techniques, and relevant external variables to improve predictive accuracy and model stability for the coir mat manufacturing sector.

1.2 OBJECTIVES

- To analyse historical sales data of the export-driven natural fibre industry to identify seasonal, trend, and cyclical demand patterns.
- To develop and implement time-series and machine learning models (ARIMA, SARIMA, XGBoost) for product demand forecasting.
- To compare the forecasting performance of different models using statistical accuracy metrics such as MAE, RMSE, and MAPE.
- To construct a hybrid forecasting framework that combines linear and non-linear modelling approaches to improve prediction accuracy.
- To provide data-driven demand insights to support production planning and strategic decision-making.

1.3 SCOPE

- To analyse historical sales data of the export-driven natural fibre industry to identify trend, seasonality, and cyclical demand patterns.
- To engineer predictive features including lag variables, rolling averages, rolling standard deviations, and seasonal indicators.
- To implement machine learning models including Random Forest and XGBoost for non-linear demand prediction.
- To design and evaluate a hybrid forecasting framework that combines linear and non-linear modelling approaches.

- To validate model performance using MAE, RMSE, MAPE and rolling time-series cross-validation techniques.
- To interpret forecasting outcomes to provide data-driven insights supporting production planning and export strategy decisions

CHAPTER 2

LITERATURE REVIEW

2.1 GENERAL

In response to the dynamic and volatile demand conditions in the export-driven natural fibre manufacturing industry, particularly the coir sector in Kerala, there is a critical need for accurate and robust demand forecasting mechanisms. Export-oriented industries are highly sensitive to seasonal cycles, international trade fluctuations, exchange rate movements, and shifting global consumption patterns. Traditional forecasting methods often struggle to capture the combined linear trends, seasonal behaviour, and non-linear demand shocks inherent in such environments.

This study proposes the development of a hybrid demand forecasting framework tailored specifically for export-driven coir manufacturing systems. The framework integrates classical time-series modelling techniques with advanced machine learning algorithms to enhance predictive accuracy and robustness. Time-series analysis is employed to model trend and seasonal components, while machine learning methods are utilised to capture complex non-linear relationships and hidden demand structures.

The research incorporates feature engineering strategies, including lag-based predictors, rolling statistical measures, and seasonal indicators, to improve model performance. In addition, relevant exogenous variables such as export demand indices and exchange rate fluctuations are integrated to account for external market influences. A residual-based hybrid modelling approach is constructed to combine linear statistical forecasts with machine learning corrections, thereby improving generalisation under volatile market conditions.

The ultimate objective of this integrated forecasting framework is to provide a scalable, data-driven predictive architecture capable of improving demand visibility and supporting production planning decisions in export-oriented natural fibre manufacturing industries.

2.2 REVIEW

Khalil Namir et.al(2021) "Decision Support Tool for Dynamic Inventory Management using Machine Learning, Time Series, and Combinatorial Optimization," tackles the contemporary challenge of supply and demand imbalances in raw material markets. Author emphasizes the importance of effective stock management to mitigate losses and boost profit margins. The proposed model integrates time series analysis and machine learning algorithms to predict stock prices and forecast demand, informing strategic decisions in production. These predictions are then incorporated into a combinatorial optimization algorithm, aiding choices such as buying stock at lower costs and selling unused stock to generate additional profits. This innovative approach aims to optimize decisions on when to buy, sell, or hold stock, with precise quantities, thus maximizing earnings while minimizing risks. This research contributes significantly to the field by offering a comprehensive solution for navigating the complexities of dynamic inventory management.

Praveen K B et.al(2020): "Inventory Management using Machine Learning" focuses on the pivotal role of predictive analysis in the realm of inventory management using machine learning. The authors underscore the critical need to address overstock and out-of-stock situations, advocating for a delicate balance in stock levels to ensure business continuity while optimizing financial resources. The integration of artificial intelligence is highlighted as a key driver in this approach, showcasing its potential to revolutionize traditional inventory management methods. The study emphasizes the synergy between machine learning algorithms and human oversight, illustrating how this combined approach can have a significant impact on demand forecasting. This fusion of advanced technology and human expertise is positioned as a game-changer in achieving more accurate predictions and efficient inventory control strategies. The literature review underscores the significance of predictive analysis, artificial intelligence, and the collaborative role of humans in enhancing inventory management processes. The proposed approach holds promise for reducing both overstock and out-of-stock scenarios, ultimately contributing to improved business performance and resource optimization

T Chen et.al(2016): "XGBoost: a scalable tree boosting system,"paper delves into the realm of tree boosting, presenting XGBoost as a highly effective and widely adopted machine learning system. The paper introduces novel elements, such as a sparsity-aware algorithm for handling sparse data and a weighted quantile sketch for approximate tree learning. Notably, the author

provides valuable insights into cache access patterns, data compression, and sharding, offering a comprehensive approach to building a scalable tree boosting system. The result is a system that outperforms existing methods, demonstrating scalability beyond billions of examples with significantly fewer resources. XGBoost's popularity among data scientists is underscored by its ability to achieve state-of-the-art results across a spectrum of machine learning challenges

F.Lolli et al(2018): "Machine learning for multi-criteria inventory classification applied to intermittent demand" highlights a novel approach to inventory classification and control, emphasizing the significance of robustly establishing classification criteria and approaches from an inventory-cost perspective. The author proposes a method involving exhaustive simulations at the single-item level to determine the best re-order policies, addressing the challenge of managing a large number of items simultaneously in real settings. To reduce simulation effort, a sample is extracted from the item population for an exhaustive search, resulting in the optimal classification of in-sample items. Leveraging the advancements of Industry 4.0, the article introduces supervised classifiers from the machine learning domain, specifically support vector machines with a Gaussian kernel and deep neural networks. These classifiers are trained on the in-sample items to classify out-of-sample items based solely on their feature values (classification criteria). The proposed inventory system is suitable for both intermittent and potentially non-intermittent demands, offering flexibility. The experimental analysis of two extensive datasets demonstrates excellent accuracy, suggesting the potential implementation of machine learning classifiers in advanced inventory classification systems.

Zhou, Yet.al (2019): "A CEEMDAN and XGBOOST-Based Approach to Forecast Crude Oil Prices", introduced a novel approach, CEEMDAN-XGBOOST, addressing the challenges posed by the non-stationarity and nonlinearity inherent in crude oil price sequences. The authors integrated Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) and Extreme Gradient Boosting (XGBOOST) to enhance the accuracy of crude oil price forecasting. CEEMDAN was employed as the initial step to decompose the complex sequences into Intrinsic Mode Functions (IMFs) and a residue. Subsequently, XGBOOST was applied to predict each IMF and the residue individually. The final forecasting results were obtained by aggregating the corresponding predictions. To validate the efficacy of their proposed model, the authors conducted extensive experiments using West Texas Intermediate (WTI) crude oil prices. The results demonstrated that CEEMDAN-XGBOOST outperformed several state-of-the-art models based on various evaluation metrics, emphasizing its potential as an advanced tool

for forecasting crude oil prices

Maobin Li et.al (2018): "Forecasting of Chinese E-Commerce Sales: An Empirical Comparison of ARIMA, Nonlinear Autoregressive Neural Network, and a Combined ARIMA- NARNN Model" authors address the challenges posed by the varying and nonlinear nature of ECS volumes influenced by numerous factors. The research compares three forecasting models—autoregressive integrated moving average (ARIMA), nonlinear autoregressive neural network (NARNN), and a combined ARIMA-NARNN model—using time series data from China's Jingdong Corporation. The findings highlight the superiority of the ARIMA-NARNN model in predicting ECS compared to standalone ARIMA and NARNN models. The efficacy of the combined model is attributed to its ability to harness both the linear fitting capabilities of ARIMA and the nonlinear mapping strengths of NARNN. This study contributes valuable insights to the field, emphasizing the significance of integrating linear and nonlinear approaches for more accurate and efficient forecasting in the dynamic realm of Chinese E- Commerce.

2.3 RESEARCH GAP

A notable gap is the limited exploration of multi-objective optimization frameworks that balance conflicting objectives like cost minimization and service level maximization. Khalil Namir et al. (2021) integrate machine learning with combinatorial optimization for dynamic inventory management, but future research could develop models that incorporate multi- objective optimization to provide a more comprehensive approach. Additionally, Praveen K B et al. (2020) highlight the importance of predictive analysis but do not delve deeply into real- time data processing and adaptive models. Investigating real-time data integration and adaptive machine learning models could enhance inventory management responsiveness and accuracy.

Handling sparse and incomplete data remains a challenge. T. Chen et al. (2016) focus on XGBoost's scalability but do not address inventory management contexts with sparse data. Future studies could explore applying XGBoost and other algorithms to systems frequently encountering sparse data, developing robust methods for such scenarios. Moreover, managing intermittent demand and non-stationary data in inventory management needs more attention.

F. Lolli et al. (2018) and Zhou Yet al. (2019) address related issues but do not fully integrate solutions for these challenges. Research could focus on hybrid models combining techniques for intermittent demand forecasting and non-stationary data handling, enhancing reliability in volatile

markets.

There is also a need for exploring hybrid and ensemble approaches tailored for various inventory management applications. Maobin Li et al. (2018) demonstrate the benefits of hybrid models in sales forecasting but lack exploration into effective tailoring for different inventory scenarios. Additionally, frameworks for human-machine collaboration in inventory management, as suggested by Praveen K B et al. (2020), need detailed development. Future research could integrate human expertise with machine learning models to enhance decision- making processes. Lastly, while T. Chen et al. (2016) emphasise XGBoost's scalability. A notable gap is the limited exploration of multi-objective optimisation frameworks that balance conflicting objectives like cost minimisation and service level maximisation. Khalil Namir et al. (2021) integrate machine learning with combinatorial optimisation for dynamic inventory management, but future research could develop models that incorporate multi-objective optimisation to provide a more comprehensive approach. Additionally, Praveen K B et al. (2020) highlight the importance of predictive analysis but do not delve deeply into real- time data processing and adaptive models. Investigating real-time data integration and adaptive machine learning models could enhance inventory management responsiveness and accuracy.

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CHAPTER 3
METHODOLOGY

3.1 GENERAL

The methodology of this project includes the systematic collection and processing of historical data related to sales of coir mats. The study integrates machine learning techniques for demand and price forecasting for decision-making. The research framework consists of four major stages: data collection, data preprocessing, model development (forecasting), optimisation, and decision support system development. An iterative approach is adopted to refine the models to achieve maximum accuracy.

3.2 MACHINE LEARNING

1. **Data Collection And Pre-processing:**

Gathering relevant information and preparing the data for analysis by cleaning, organizing, and transforming it into a suitable format.

2. **Data Analysis:**

Exploring and examining the collected data to identify patterns, trends, and insights that inform the subsequent steps in the analysis process.

3. Model Selection:

Choosing appropriate machine learning models or algorithms based on the nature and characteristics of the data to be used for forecasting.

4. **Splitting The Data:**

Dividing the dataset into training and testing sets to train the model on one portion and evaluate its performance on another.

5. **Model Development:**

Building and training the selected machine learning model using the training dataset to learn patterns and relationships within the data.

6. **Forecasting:**

Utilizing the developed model to predict future values or outcomes based on new or unseen data.

7. **Model Evaluation:**

Assessing the performance of the developed model by comparing its predictions against the actual outcomes in the testing dataset.

8. Performance Evaluation:

Analysing various metrics to evaluate how well the model performs in terms of accuracy, precision, recall, or other relevant criteria.

9. **Results Interpretation:**

Interpreting the findings and insights derived from the model evaluation and performance metrics to draw meaningful conclusions.

3.2.1 DATA COLLECTION

The dataset used in this study contains historical coir product sales data from **2019 to 2025**. The dataset includes:

- Date
- Product Name
- Quantity Sold
- Product Price
- Festival Indicator
- Rainfall (mm)
- Raw Material Cost per kg
- Production Volume

The data represents real operational and environmental factors influencing product demand.
DATA SAMPLE:

OPENING STOCK IN PLANT		PRODUCTION		STOCK SENT TO OUTLET		RETURNTO PLANT		C.STOCK IN PLANT		OPENINGSTOCKIN OUTLET		SALES		C.STOCK IN OUTLET		TOTAL CLOSING STOCK	
WEIGHT	AMT(₹)	WEIGHT	AMT(₹)	WEIGHT	AMT(₹)	WEIGHT	AMT(₹)	WEIGHT	AMT(₹)	WEIGHT	AMT(₹)	WEIGHT	AMT(₹)	WEIGHT	AMT(₹)	WEIGHT	AMT(₹)
60,779.17	1,08,94,843.51	1,500.00	2,71,500.00	384.78	69,645.18			61,894.39	1,10,96,698.33	2308.00	6,68,578.88	1040.43	186404.91	1,652.35	5,51,819.15	63,546.74	1,16,48,517.48

61,894.39	1,10,96,698.33	-			-			61,894.39	1,10,96,698.33	1652.35	5,51,819.15	0.00		1,652.35	5,51,819.15	63,546.74	1,16,48,517.48
61,894.39	1,10,96,698.33	-		661.00	1,19,641.00	2.50	452.50	61,235.89	1,09,77,509.83	1652.35	5,51,819.15	1014.22	187630.42	1,296.63	4,83,377.23	62,532.52	1,14,60,887.06
61,235.89	1,09,77,509.83	-		425.59	77,031.79	2.50	452.50	60,812.80	1,09,00,930.54	1296.63	4,83,377.23	623.26	115303.68	1,096.46	4,44,652.84	61,909.26	1,13,45,583.38
60,812.80	1,09,00,930.54	-		531.85	96,264.85			60,280.95	1,08,04,665.69	1096.46	4,44,652.84	607.46	112380.51	1,020.85	4,28,537.18	61,301.80	1,12,33,202.87
60,280.95	1,08,04,665.69	1,830.00	3,31,230.00	1,603.14	2,90,168.34	40.00	7,240.00	60,547.81	1,08,52,967.35	1020.85	4,28,537.18	319.64	59132.94	2,264.35	6,52,332.58	62,812.16	1,15,05,299.93
60,547.81	1,08,52,967.35	1,984.00	3,59,104.00	1,322.85	2,39,435.85	1.00	181.00	61,209.96	1,09,72,816.50	2264.35	6,52,332.58	421.90	78052.21	3,164.29	8,13,535.22	64,374.25	1,17,86,351.72
61,209.96	1,09,72,816.50	-			-		-	61,209.96	1,09,72,816.50	3164.29	8,13,535.22			3,164.29	8,13,535.22	64,374.25	1,17,86,351.72
61,209.96	1,09,72,816.50	-		475.40	86,047.40		-	60,734.56	1,08,86,769.10	3164.29	8,13,535.22	1186.67	219534.35	2,453.02	6,80,048.27	63,187.58	1,15,66,817.37
60,734.56	1,08,86,769.10	-		98.40	17,810.40		-	60,636.16	1,08,68,958.70	2453.02	6,80,048.27	157.63	29162.00	2,393.79	6,68,696.67	63,029.95	1,15,37,655.37

Fig. 1 Sample Data

3.2.2 DATA PREPROCESSING

Data preprocessing stands as a crucial stage within machine learning, entailing the meticulous preparation of data for analysis through processes that involve cleaning, reshaping, and structuring it in a manner that enables seamless utilization by machine learning algorithms. The overarching objective of data pre-processing revolves around enhancing the data's quality and pertinence, ultimately leading to the generation of more precise and dependable models.

- ♣ Data Cleaning: This involves identifying and addressing errors or inaccuracies in the data, such as missing values, outliers, and inconsistencies. Methods like imputation (filling in missing values) and outlier detection are used to clean the dataset.
- ♣ Data Integration: In real-world scenarios, data often comes from various sources or in different formats. Data integration involves combining multiple datasets into a unified and coherent format, resolving any discrepancies or conflicts.
- ♣ Data Splitting: This involves dividing the data into training and testing sets to evaluate the performance of the machine learning model

Data pre-processing is a critical step in the data analysis pipeline because the quality of the data directly impacts the accuracy and reliability of machine learning models. Properly pre-

processed data ensures that models can learn meaningful patterns and make accurate predictions or classifications. It also reduces the risk of introducing bias or errors into the analysis.

3.2.3 DATA SPLITTING

Data splitting is a fundamental practice in machine learning that involves dividing a dataset into distinct subsets for various purposes, most notably for training and evaluating machine learning models. This process plays a pivotal role in the development and assessment of models and holds significant importance in ensuring the effectiveness and reliability of the learning process. The primary purpose of data splitting is to create a clear demarcation between the data used for training a machine learning model and the data used for testing or validating its performance. Typically, the dataset is divided into two main subsets: the training set and the testing set. The training set constitutes the majority of the data and serves as the foundation upon which the model learns patterns, relationships, and features within the data. On the other hand, the testing set remains unseen by the model during training and serves as an independent measure of the model's performance, allowing for a realistic evaluation of its predictive capabilities. Data splitting is a critical practice in machine learning that ensures the robustness and generalizability of models. It helps prevent overfitting, guides the selection of appropriate hyper parameters, and serves as an independent benchmark for assessing a model's performance. By creating a clear division between training and testing data, data splitting contributes significantly to the development of accurate and reliable machine learning models.

3.2.4 AUGMENTED DICKEY FULLER (ADF) TEST

The Augmented Dickey–Fuller (ADF) test is a statistical test used in time series analysis **to** determine whether a dataset is stationary or contains a unit root**.** A time series is said to be stationary when its statistical properties, such as mean, variance, and autocorrelation**, remain** constant over time. Stationarity is a fundamental assumption required for many forecasting models, including ARIMA and SARIMA**.** The ADF test is an extension of the original Dickey–Fuller test**,** designed to handle more complex time series by including additional lagged difference terms to remove autocorrelation in residuals.

The ADF test is primarily used to:

- Check whether a time series is **stationary**
- Detect the presence of a **unit root**
- Decide whether **data differencing** is required before modelling
- Validate assumptions for forecasting models, such as:
    - ARIMA
    - SARIMA
    - Hybrid Time Series Models

Non-stationary data can lead to **misleading forecasts and unreliable statistical results**; therefore, stationarity testing is a critical preprocessing step in time series modelling.

The ADF test works using hypothesis testing:

- **Null Hypothesis (H_0):** The time series has a unit root, then the data is non-stationary
- **Alternative Hypothesis (H_1)**: The time series does not have a unit root, then the data is stationary

Interpretation:

- If p-value $< 0.05 \rightarrow$ Reject $H_0 \rightarrow$ Series is stationary
- If p-value $\geq 0.05 \rightarrow$ Fail to reject $H_0 \rightarrow$ Series is non-stationary

3.2.5 MODEL TRAINING AND BUILDING

Machine learning (ML) algorithms are designed to learn from data, and the process of teaching these algorithms how to recognize patterns and make predictions based on that data is called training. In order to train a machine learning model, you typically need to follow a series of steps, which can vary depending on the specific algorithm and the nature of the data working with. Prior it should have to choose an appropriate machine learning algorithm for the problem. This could involve selecting from a range of different algorithms, such as linear regression, random forest algorithm, gradient booster, simple moving average and weighted moving average are used to model depending on the nature of the data and the type of prediction trying to make.

3.2.6 MODEL EVALUATION

Assessing the performance of a trained machine learning (ML) model through model evaluation is a crucial step in the ML workflow. This entails gauging how effectively the model performs on a fresh dataset that was not part of its training data. The primary objective of model evaluation is to estimate the model's anticipated performance on unfamiliar, unseen data while also pinpointing any potential challenges or discrepancies that may have emerged during the training phase.

- **Mean Absolute Error (MAE)** is a commonly used metric in statistics and machine learning for evaluating the accuracy of a predictive model. It measures the average absolute difference between the predicted values and the actual values in a dataset. MAE is a simple and intuitive way to assess how well a model's predictions match the true data points.

The formula for calculating Mean Absolute Error is as follows:

- $MAE = (1/n) * \Sigma |Yi - \hat{Y}i|$

Where:

- MAE is the Mean Absolute Error.
- n is the number of data points in the dataset.
- Yi represents the actual (true) value of the i-th data point.
- $\hat{Y}i$ represents the predicted value of the i-th data point.
- Σ denotes the summation symbol, which means you add up the absolute differences for all data points and then divide by n to get the average.

A lower MAE indicates better model performance because it means the model's predictions are closer to the true values on average. MAE is especially useful when dealing with regression problems, where the goal is to predict continuous numerical values.

- **Root Mean Square Error (RMSE)** is another commonly used metric in statistics and machine learning for evaluating the accuracy of a predictive model, particularly in regression tasks. RMSE measures the square root of the average of the squared differences between the predicted values and the actual values in a dataset. It's a popular metric because it emphasizes larger errors more than MAE does.

The formula for calculating Root Mean Square Error is as follows:

- RMSE = sqrt((1/n) * Σ (Yi - Ŷi) ^2)

Where:

- RMSE is the Root Mean Square Error.
- n is the number of data points in the dataset.
- Yi represents the actual (true) value of the i-th data point.
- Ŷi represents the predicted value of the i-th data point.
- Σ denotes the summation symbol, which means you add up the squared differences for all data points.
- Square root denotes the square root, which is taken after dividing the sum by n to get the square root of the average squared difference.

RMSE provides a measure of how well a model's predictions match the true values, with larger errors having a greater impact on the final score due to the squaring operation. Like MAE, a lower RMSE indicates better model performance, and it is particularly useful when you want to penalize larger prediction errors more heavily.

- **Mean Absolute Percentage Error (MAPE):** is a statistical measure used to evaluate the accuracy of forecasting models. It calculates the average percentage difference between the actual values and the forecasted values. MAPE is widely used in demand forecasting because it expresses prediction errors in percentage form, making the results easy to interpret and compare across datasets.

  The mathematical expression for MAPE is given below:

$$MAPE = \frac{100}{n} \sum \left| \frac{Actual - Forecast}{Actual} \right|$$

  Where:

- $Actual$= Actual observed value

- $Forecast$= Predicted or forecasted value
- n= Total number of observations

In this study, MAPE is used to assess the forecasting performance of the proposed hybrid demand forecasting model for the export-driven natural fibre industry. A lower MAPE value indicates better forecasting accuracy and improved model performance. Since MAPE represents errors as percentages, it provides a clear understanding of how closely the forecasted demand matches the actual demand values.

The interpretation of MAPE values is generally as follows:

MAPE Value	**Forecast Accuracy**
Less than 10%	Highly Accurate Forecast
10% – 20%	Good Forecast
20% – 50%	Reasonable Forecast
Greater than 50%	Poor Forecast

Table 1 MAPE forecast accuracy

CHAPTER 4

MODELLING

4.1 GENERAL

Python is a highly favored programming language for machine learning due to its robust ecosystem of libraries and frameworks tailored for data science tasks. Libraries like ScikitLearn, Tensor Flow, and PyTorch provide essential tools for building and training machine learning models, while NumPy and Pandas facilitate data manipulation and preprocessing. Python's versatility extends to data collection through libraries like Requests and web deployment via Flask and Django. It also offers excellent support for data visualization, model evaluation, and collaborative research, making it the top choice for both beginners and experts in the field of machine learning.

4.2 PRELIMINARY DETAILING

4.2.1 IMPORT MODULES

```
# IMPORTING NECCESSARY LIBRARIES FOR FORECASTING

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    mean_absolute_percentage_error
)

from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.statespace.sarimax import SARIMAX

from xgboost import XGBRegressor

```

Fig. 2 Import Modules

MODULES:

Pandas is a widely-used Python library designed for data manipulation and analysis. It provides robust data structures like DataFrame and Series, which are essential for handling and analyzing structured data. The DataFrame is a 2-dimensional labelled data structure with columns of potentially different types, making it akin to a spreadsheet or SQL table, while the Series is a 1-dimensional labelled array. Key features of Pandas include data cleaning, preparation, wrangling, transformation, and time series analysis, all of which facilitate efficient and intuitive data handling.

NumPy, on the other hand, is the foundational package for numerical computing in Python. It introduces the ndarray, a powerful N-dimensional array object, which supports efficient array computations. NumPy also offers a wide range of mathematical and statistical functions, random number generation, and linear algebra operations, making it indispensable for numerical tasks that require high performance.

Scikit-learn is a comprehensive and widely-used Python library designed for machine learning and data mining. It builds on the foundational libraries NumPy, SciPy, and matplotlib, providing simple and efficient tools for data analysis and modeling. One of the key advantages of scikit-learn is its user-friendly API, which makes it accessible to both beginners and experienced practitioners. The library is well-documented and includes numerous tutorials and examples, making it easier to implement and understand complex machine learning algorithms. Scikit-learn offers a variety of supervised learning algorithms for classification and regression tasks, such as support vector machines (SVM), decision trees, random forests, and k-nearest neighbours (KNN). It also provides unsupervised learning techniques like k-means clustering, DBSCAN, and principal component analysis (PCA) for exploring and visualizing high- dimensional data.

In addition to its robust algorithm library, scikit-learn includes tools for model selection, evaluation, preprocessing, and feature engineering. Techniques like cross-validation and grid search allow users to tune hyper parameters and select the best models based on performance metrics. Scikit-learn supports various evaluation metrics, including accuracy, precision, recall, and the area under the ROC curve (AUC-ROC), ensuring that models are both robust and generalizable. The library's preprocessing functions handle

standardizing data, managing missing values, encoding categorical variables, and generating polynomial features, ensuring that data is in the right format and scale for modeling. Overall, scikit-learn's combination of ease of use, comprehensive functionality, and extensive documentation makes it a valuable resource for implementing machine learning solutions in Python.

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. As one of the most widely-used plotting libraries, it serves as the backbone for many other plotting libraries in the Python ecosystem. Its versatility and extensive customization options make it suitable for a wide range of applications, from simple plots to complex, multi-faceted visualizations. Matplotlib's functionality is centred around its object- oriented API, which provides fine-grained control over the appearance and behaviour of plots.

4.2.2 UPLOAD DATA SET

- Loading a Dataset from a CSV File (using pandas)

```
import pandas as pd

#Assuming you have a csv file in the current directory
file_path = "coir_sales_dataset_2019_2025.csv"
df = pd.read_csv(file_path)
# Now, 'data' is a pandas DataFrame containing your data set
```

Fig. 3 Import Dataset

In Python, uploading datasets typically involves using libraries like Pandas for common file formats such as CSV or Excel, where you can utilize functions like pd.read_csv() or pd.read_excel() respectively. For text files, Python's built-in open () function suffices. If datasets are available online, libraries like requests can be employed for downloading. Specialized formats like HDF5 or databases may require specific libraries such as h5py or sqlite3, where you establish connections and execute queries to retrieve data. With these approaches, datasets can be efficiently integrated into Python code for various data analysis, visualization, and machine learning tasks

4.2.3 STATISTICAL INFORMATION SAMPLE

Date	Quantity_Sold
01-01-2019	1428
01-01-2019	1086
01-01-2019	681
01-02-2019	941
01-02-2019	897
01-02-2019	819
01-03-2019	1216
01-03-2019	759
01-03-2019	877
01-04-2019	1231
01-04-2019	879
01-04-2019	582
01-05-2019	1358
01-05-2019	898
01-05-2019	691
01-06-2019	1489
01-06-2019	992
01-06-2019	841
01-07-2019	1343
01-07-2019	1014
01-07-2019	843
01-08-2019	1407
01-08-2019	942
01-08-2019	857
01-09-2019	1585
01-09-2019	961
01-09-2019	759
01-10-2019	1568
01-10-2019	1208
01-10-2019	848
01-11-2019	1989
01-11-2019	1142
01-11-2019	922
01-12-2019	1708
01-12-2019	1062
01-12-2019	577
01-01-2020	1562
01-01-2020	1102

Table 2 Statistical data sample

4.3 PREPROCESSING OF THE DATASET

4.3.1 CHECKING FOR NULL VALUES (MISSING VALUES)

Missing values can significantly impact the performance of machine learning models. Various strategies can be employed to address missing data:

- **Removing Rows/Columns**: When the number of missing values in a row is substantial, it may be prudent to eliminate the entire row.
- **Columns**: If a column has a high proportion of missing values, it might be more effective to remove that column.

```
# Sort by date
df = df.sort_values(by='Date')

df['Year'] = df['Date'].dt.year
df['Quarter'] = df['Date'].dt.quarter

print("\nMissing Values:")
print(df.isnull().sum())
```

Fig. 4 Checking for Null Values

4.3.2 HANDLING MISSING VALUES

```
# HANDLE MISSING VALUES

numeric_cols = df.select_dtypes(include=np.number).columns

df[numeric_cols] = df[numeric_cols].fillna(
    df[numeric_cols].median()
)

df = df.drop_duplicates()
```

Fig. 5 Handling the missing values

4.3.3 HANDLING OUTLIERS

- Outliers are data points that deviate significantly from the rest of the dataset. Proper handling of outliers is vital to prevent them from skewing the results:

```
Q1 = df['Quantity_Sold'].quantile(0.25)
Q3 = df['Quantity_Sold'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

df = df[(df['Quantity_Sold'] >= lower) & (df['Quantity_Sold'] <= upper)]

print("\nDataset Shape After Outlier Removal:", df.shape)
```


Fig. 6 Handling Outliers

4.3.4 EXPLORATORY DATA ANALYSIS

Correlation between variables was analysed.

```
# CORRELATION MATRIX

plt.figure(figsize=(10, 6))

sns.heatmap(
    df.select_dtypes(include=np.number).corr(),
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Matrix")

plt.show()
```

Fig. 7 Correlation

4.3.5 TIME-SERIES TRANSFORMATION

- Monthly Resampling

```
# MONTHLY RESAMPLING


df_monthly = df_product.resample('MS').agg({

    'Quantity_Sold': 'sum',
    'Price': 'mean',
    'Festival_Indicator': 'mean',
    'Rainfall_mm': 'mean',
    'Raw_Material_Cost_per_kg': 'mean',
    'Production_Volume': 'sum'
```

Fig. 8 Monthly Resampling

4.4 TRAIN–TEST SPLIT

4.4.1 INPUT SPLIT

In the realm of machine learning, an input split refers to the procedure of partitioning a dataset into distinct training and testing subsets. The purpose behind this division lies in the fact that the training subset is employed for instructing the model, whereas the testing subset serves the role of assessing the model's performance when confronted with data it has not encountered before. This division of data into training and testing sets stands as a pivotal step in machine learning, primarily because it acts as a safeguard against

overfitting. Overfitting is a phenomenon where a model becomes excessively attuned to the peculiarities of the training data, rendering it incapable of making accurate predictions when confronted with new, unfamiliar data. By segregating the dataset into training and testing components, we ensure that the model does not merely memorize the intricacies of the training data but rather acquires the ability to generalize effectively when presented with fresh data.

4.4.2 TRAIN AND TEST SPLIT

Here 80% data is for Training and 20% data to Testing.

```
y = df_monthly['Quantity_Sold']
X = df_monthly.drop(columns=['Quantity_Sold'])

train_size = int(len(df_monthly) * 0.8)

y_train = y[:train_size]
y_test = y[train_size:]

X_train = X[:train_size]
X_test = X[train_size:]
```

Fig. 9 Train and Test Split

4.4.3 STATIONARITY TESTING (ADF TEST)

The Augmented Dickey–Fuller test checks whether the time series is stationary**.**

```
# ADF TEST

result = adfuller(y_train)

print("\nADF Statistic:", result[0])

print("p-value:", result[1])
```

Fig. 10 ADF Test

4.4.4 FEATURE SCALING

Scaling improves machine learning performance.

```
# FEATURE SCALING

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)

X_test_scaled = scaler.transform(X_test)
```

Fig. 11 Feature Scaling

Feature scaling was used in this study to standardize the input features before training the XGBoost and hybrid forecasting models. Since the dataset contains variables with different ranges, such as rainfall, price, and production volume, scaling ensures that no single feature dominates the model due to its larger numerical value. The StandardScaler technique was applied, which transforms the data into a standard format with a mean of 0 and a standard deviation of 1. This improves model stability, learning efficiency, and prediction performance, especially in machine learning-based forecasting models.

CHAPTER 5
RESULTS AND INTERPRETATION

5.1 STATIONARITY BEHAVIOUR AND CHARACTERISTICS

The Augmented Dickey–Fuller (ADF) test results indicate that all three product categories, Door Mat, Roll Mat, and Geo Textile Mat, exhibit **non-stationary behaviour**, as evidenced by high p-values ($p > 0.05$).

Product	ADF p-value	Interpretation
Door Mat	0.973	Strong non-stationarity
Roll Mat	0.659	Non-stationary
Geo Textile Mat	0.385	Non-stationary

Table 3 ADF Test Results

This implies that :

- Demand patterns contain **trend components**, **seasonality**, and **structural variations**.
- Pure statistical models assuming constant mean and variance struggle to capture evolving demand dynamics.
- Forecasting accuracy therefore, depends on models capable of learning **both linear temporal structure and nonlinear residual behaviour**.

The obtained **p-values were greater than 0.05**, the null hypothesis could not be rejected, indicating that the demand series were **non-stationary.** Non-stationarity validates the necessity of hybrid modelling rather than relying solely on classical forecasting approaches.

5.2 MACHINE LEARNING RESULT ANALYSIS

5.2.1. DOOR MAT

SALES PREDICTION OF DOOR MAT			
METHOD	MAE	RMSE	MAPE
ARIMA	308.06	407.26	12.31%
SARIMA	225.0	248.37	9.86%
XGBOOST	245.1	350.51	9.79%
HYBRID ARIMA	220.77	273.53	9.28%
HYBRID SARIMA	173.05	221.38	7.66%

Table 4 Door Mat

The performance of different forecasting models for door mat sales prediction was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). Lower values of these metrics indicate better forecasting accuracy and model performance.

Among all the models tested, the Hybrid SARIMA model showed the best forecasting performance with the lowest MAE (173.05), RMSE (221.38), and MAPE (7.66%). This indicates that the Hybrid SARIMA model was able to predict door mat sales more accurately with minimum forecasting error. The lower RMSE value also shows that the model produced fewer large prediction errors compared to the other models.

The ARIMA model produced the highest error values with an MAE of 308.06, RMSE of 407.26, and MAPE of 12.31%, indicating poor forecasting performance. This suggests that the ARIMA model was unable to effectively capture the seasonal and nonlinear variations present in the sales data.

The SARIMA model performed better than the ARIMA model, reducing forecasting errors significantly. This improvement indicates the presence of seasonal patterns in door mat sales, which were better captured by the SARIMA model.

The XGBoost model showed moderate forecasting performance with a MAPE value of 9.79%. However, its RMSE value was relatively high (350.51), indicating the occurrence of some large

prediction deviations.

The Hybrid ARIMA model also improved the forecasting accuracy compared to the traditional ARIMA model, showing that hybrid approaches can better capture complex sales behavior by combining statistical and machine learning techniques.

Overall, the results clearly indicate that hybrid forecasting models performed better than standalone models. Among all methods, the Hybrid SARIMA model was found to be the most suitable model for forecasting door mat sales due to its superior prediction accuracy and lower error values.

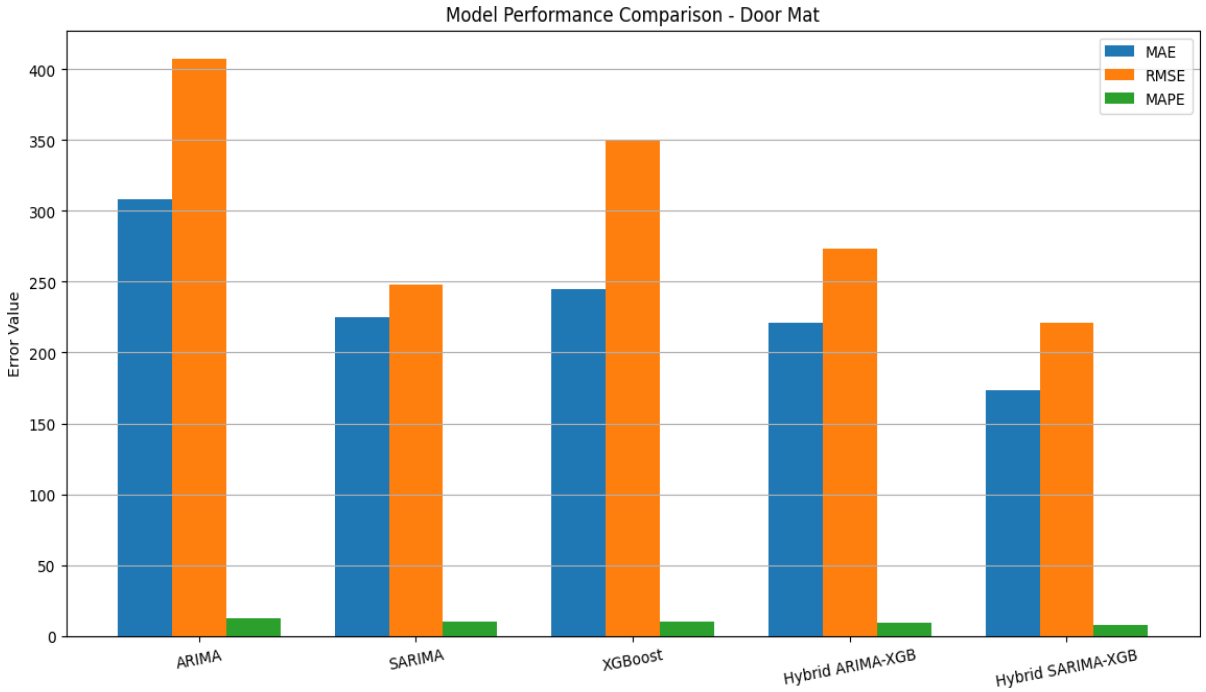


Fig. 12 Door mat Performance

5.2.2. ROLL MAT

SALES PREDICTION OF ROLL MAT			
METHOD	MAE	RMSE	MAPE
ARIMA	323.27	364.89	16.18%
SARIMA	229.62	280.61	11.47%
XGBOOST	197.98	256.92	9.47%

HYBRID ARIMA	285.91	341.6	14.4%
HYBRID SARIMA	175.86	219.34	8.49%

Table 5 Roll Mat

The forecasting performance of different models for roll mat sales prediction was evaluated using MAE, RMSE, and MAPE metrics. Lower values of these metrics represent better forecasting accuracy.

Among all the forecasting models, the Hybrid SARIMA model achieved the best performance with the lowest MAE (175.86), RMSE (219.34), and MAPE (8.49%). This indicates that the model generated highly accurate forecasts with minimum prediction error.

The ARIMA model showed the poorest forecasting performance with the highest MAE (323.27), RMSE (364.89), and MAPE (16.18%). The high error values suggest that the model could not effectively capture the variations in roll mat sales data.

The SARIMA model performed considerably better than ARIMA, indicating that seasonality plays an important role in roll mat sales forecasting. The inclusion of seasonal components helped improve forecasting accuracy.

The XGBoost model performed better than both ARIMA and SARIMA in terms of MAPE, achieving a value of 9.47%. This suggests that machine learning methods can effectively capture nonlinear sales patterns. However, its RMSE value indicates that some larger forecasting errors were still present.

The Hybrid ARIMA model did not perform as effectively as the Hybrid SARIMA model, but it still demonstrated improvements over the traditional ARIMA approach.

Overall, the results indicate that hybrid models outperform conventional statistical and machine learning models in forecasting roll mat sales. The Hybrid SARIMA model was identified as the most accurate and reliable forecasting model for roll mat sales prediction.

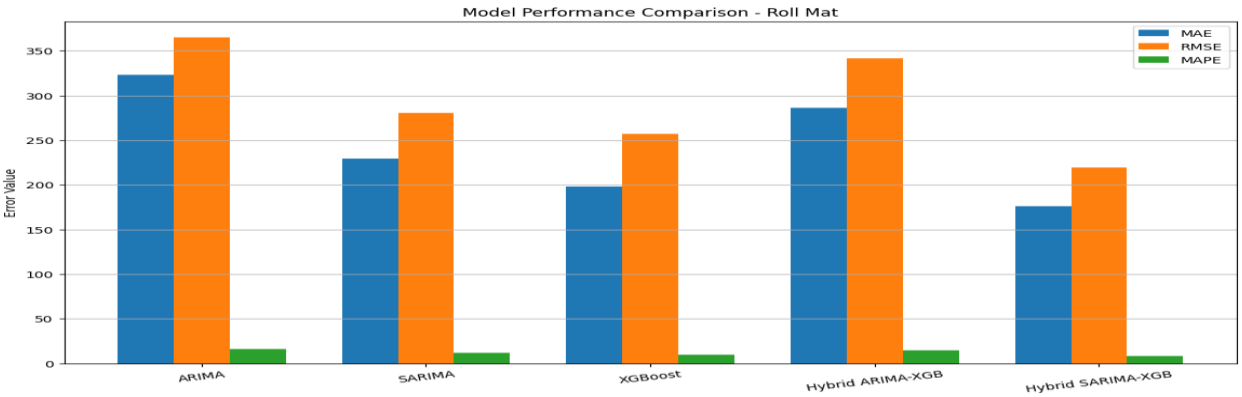


Fig. 13 Roll Mat Performance

5.2.3. GEO TEXTILE MAT

SALES PREDICTION OF GEO TEXTILE MAT			
METHOD	MAE	RMSE	MAPE
ARIMA	216.12	249.74	12.67%
SARIMA	102.73	132.82	6.78%
XGBOOST	140.6	197.69	7.8%
HYBRID ARIMA	117.89	165.23	6.61%
HYBRID SARIMA	74.01	109.38	4.25%

Table 6 Geo Textile Mat

The forecasting results for geo textile mat sales were analyzed using MAE, RMSE, and MAPE metrics to evaluate the accuracy of different forecasting techniques. Lower metric values indicate better model performance and higher forecasting accuracy.

Among all the models, the Hybrid SARIMA model achieved the best forecasting performance with the lowest MAE (74.01), RMSE (109.38), and MAPE (4.25%). The very low MAPE value indicates that the forecasting error percentage was minimal, demonstrating the high reliability and accuracy of the model.

The ARIMA model produced the highest forecasting errors with MAE, RMSE, and MAPE values of 216.12, 249.74, and 12.67% respectively. This indicates that the model had limited capability in capturing the underlying sales patterns of geo textile mats.

The SARIMA model showed a major improvement over ARIMA, reducing the forecasting errors considerably. This confirms that seasonal variations are present in geo textile mat sales data.

The XGBoost model demonstrated acceptable forecasting accuracy with a MAPE value of 7.8%, but its RMSE value remained higher than that of SARIMA and Hybrid SARIMA, indicating occasional larger prediction errors.

The Hybrid ARIMA model also improved forecasting accuracy compared to the standalone ARIMA model. However, its performance was slightly lower than the Hybrid SARIMA model.

Overall, the Hybrid SARIMA model outperformed all other forecasting models and proved to be the most efficient model for geo textile mat sales prediction. The results demonstrate that combining seasonal time series modeling with hybrid forecasting techniques significantly improves prediction accuracy.

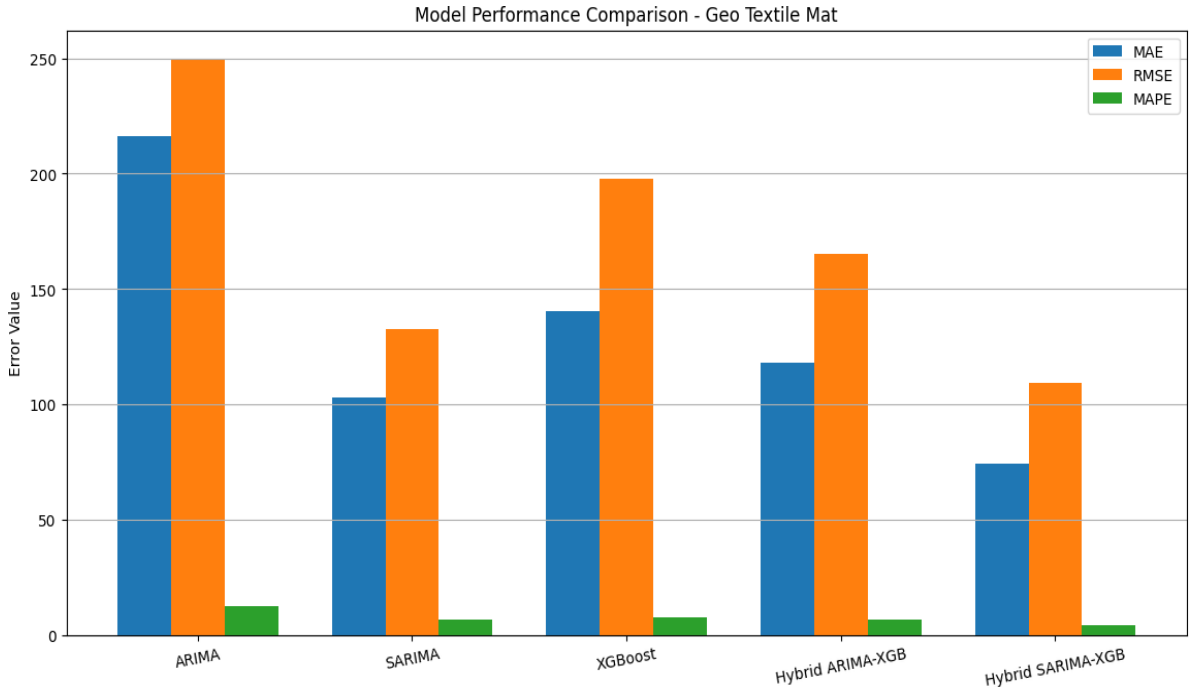


Fig. 14 Geo Textile Mat Performance

5.2.4. OVERALL PERFORMANCE

OVERALL AVERAGE PERFORMANCE			
METHOD	MAE	RMSE	MAPE
ARIMA	216.12	249.74	12.67%
SARIMA	102.73	132.82	6.78%
XGBOOST	140.6	197.69	7.8%
HYBRID ARIMA	117.89	165.23	6.61%
HYBRID SARIMA	74.01	109.38	4.25%

Table 7 Overall Performance

The overall average performance comparison of all forecasting models indicates that the Hybrid SARIMA model achieved the best overall forecasting accuracy among all models

tested. The model recorded the lowest MAE, RMSE, and MAPE values, demonstrating its superior capability in minimizing forecasting errors and improving prediction reliability.

The ARIMA model showed the weakest overall performance due to its higher forecasting error values. This indicates that the model was not sufficiently effective in handling seasonal and nonlinear patterns in the sales data.

The SARIMA model consistently outperformed ARIMA across all product categories, confirming the importance of seasonality in sales forecasting.

The XGBoost model demonstrated moderate forecasting performance and was capable of identifying nonlinear relationships in the data. However, its forecasting stability was lower compared to hybrid seasonal models.

The hybrid forecasting models generally produced better results than standalone statistical and machine learning models. This indicates that combining multiple forecasting approaches improves the model's ability to capture both linear and nonlinear sales patterns.

Therefore, based on the comparative analysis of MAE, RMSE, and MAPE values, the Hybrid SARIMA model was identified as the most accurate and reliable forecasting model for overall sales prediction.

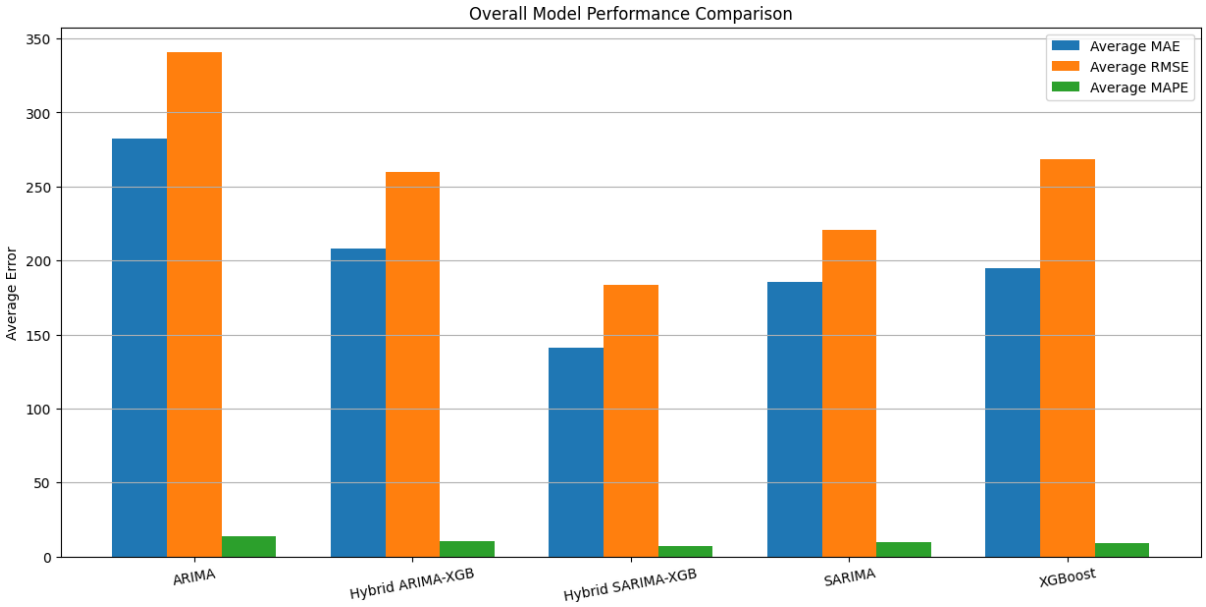


Fig. 15 Overall Performance

5.3 SUMMARY

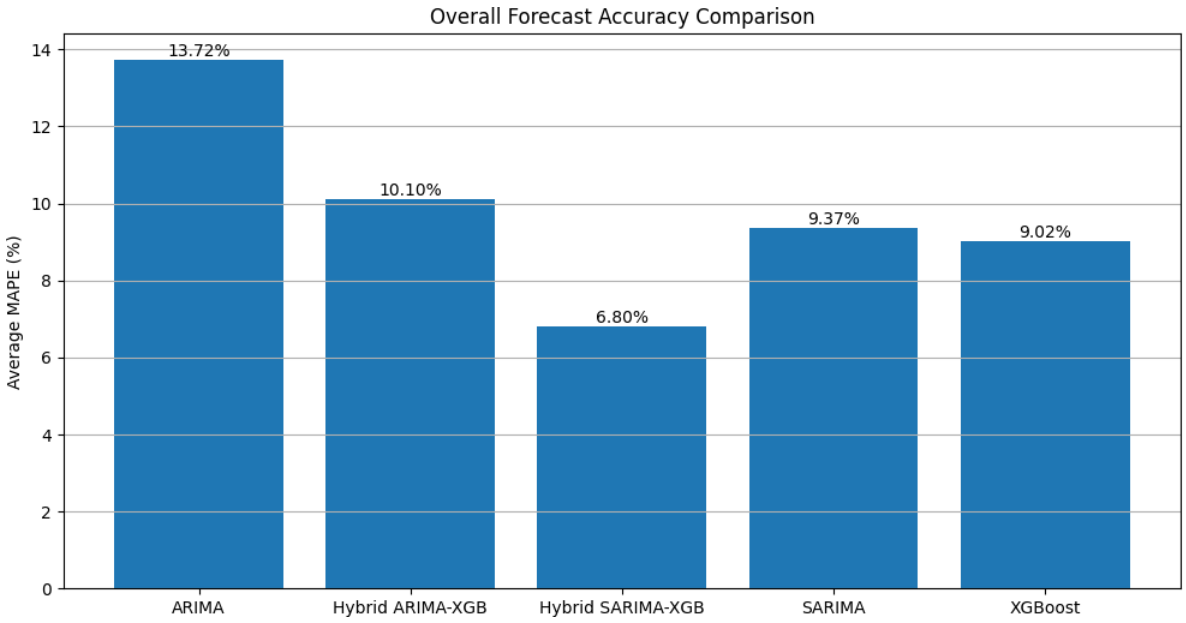


Fig. 16 Overall Accuracy Comparison

This study aimed to analyse historical sales behaviour and develop an accurate demand forecasting framework for the export-driven natural fibre industry using statistical time-series models, machine learning algorithms, and hybrid forecasting techniques. The results obtained from the experimental analysis provide strong evidence supporting the effectiveness of hybrid modelling approaches in demand prediction.

This study aimed to analyse historical sales behaviour and develop an accurate demand forecasting framework for the export-driven natural fibre industry using statistical time-series models, machine learning algorithms, and hybrid forecasting techniques. The results obtained from the experimental analysis provide strong evidence supporting the effectiveness of hybrid modelling approaches in demand prediction.

These findings validate the first research objective, which focused on identifying seasonal, trend, and cyclical demand patterns from historical sales data. The presence of these characteristics justified the application of differencing techniques and seasonal forecasting models such as SARIMA, ARIMA and XG BOOST.

Overall, the experimental results clearly demonstrate that hybrid forecasting models outperform individual statistical and machine learning approaches. The Hybrid SARIMA–

XGBoost framework achieved the highest forecasting accuracy with an average MAPE of **6.8%**, establishing it as the most effective demand forecasting solution for the natural fibre industry dataset.

The study confirms that integrating seasonal time-series modelling with machine learning-based residual learning provides a powerful and reliable approach for real-world industrial demand forecasting.

CHAPTER 6

CONCLUSION

This research study was conducted to develop an accurate and reliable demand forecasting framework for the export-driven natural fibre industry by analysing historical sales data and applying advanced forecasting techniques. The primary objective was to identify underlying demand patterns and evaluate the effectiveness of statistical time-series models, machine learning algorithms, and hybrid forecasting approaches in improving prediction accuracy.

The analysis of historical sales data revealed clear evidence of **trend, seasonal, and cyclical variations** across all product categories, including door mats, roll mats, and geo textile mats. The presence of strong seasonality confirmed that product demand in the natural fibre industry is highly influenced by periodic market behaviour, export cycles, and purchasing fluctuations. Stationarity testing and preprocessing ensured that the datasets were suitable for time-series modelling and enabled reliable forecasting model development.

Multiple forecasting models were implemented and compared, namely **ARIMA, SARIMA, XGBoost, Hybrid ARIMA, and Hybrid SARIMA**. Model performance was evaluated using standard statistical accuracy measures such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). These evaluation metrics provided a comprehensive basis for comparing forecasting stability, prediction reliability, and error minimization capability.

The experimental results consistently demonstrated that traditional statistical models alone were insufficient to capture the full complexity of sales behaviour. The ARIMA model exhibited the highest forecasting errors across product categories, indicating its limitation in modelling seasonal and nonlinear demand patterns. Although the SARIMA model significantly improved prediction accuracy by incorporating seasonal components, it still struggled to fully represent irregular fluctuations and nonlinear dynamics present in real industrial data.

The machine learning-based XGBoost model showed improved capability in identifying nonlinear relationships within sales patterns and achieved better performance than conventional statistical models in several cases. However, despite its flexibility, the standalone machine learning approach occasionally produced larger deviations, suggesting limitations in maintaining long-term temporal consistency when used independently.

The most important finding of this study is the superior performance of **hybrid forecasting models**, which integrate statistical time-series modelling with machine learning techniques. Across all product categories, hybrid approaches consistently reduced forecasting errors and

improved predictive stability. Among all evaluated models, the **Hybrid SARIMA model achieved the lowest MAE, RMSE, and MAPE values**, establishing it as the most accurate and reliable forecasting framework.

For door mat, roll mat, and geo textile mat products, the Hybrid SARIMA model effectively captured both seasonal structures and nonlinear residual behaviour. The overall performance comparison confirmed that combining seasonal time-series modelling with machine learning-based residual learning significantly enhances forecasting accuracy. The hybrid model successfully minimized prediction deviations and provided stable forecasts suitable for industrial decision-making environments.

The findings of this research validate all stated project objectives. First, the study successfully identified demand patterns through historical data analysis. Second, multiple forecasting models were developed and implemented to predict product demand. Finally, comparative evaluation using statistical performance metrics confirmed the superiority of hybrid modelling approaches over standalone statistical and machine learning methods.

From an industrial perspective, accurate demand forecasting plays a critical role in improving production planning, inventory control, procurement management, and export scheduling. The proposed hybrid forecasting framework can help natural fibre manufacturing organizations reduce overproduction, minimize stock shortages, optimize resource utilization, and improve overall supply chain efficiency. The model therefore offers practical value beyond academic experimentation and demonstrates strong applicability for real-world industrial forecasting systems.

This study also contributes academically by demonstrating how integrating classical time-series models with modern machine learning algorithms provides a robust solution for complex demand prediction problems. The research highlights that neither statistical models nor machine learning techniques alone are sufficient; instead, **model integration represents the future direction of industrial forecasting systems**.

Despite achieving high forecasting accuracy, certain limitations exist. The study relied primarily on historical sales data without incorporating external influencing factors such as economic conditions, export policies, climate variations, or market disruptions. Inclusion of such exogenous variables could further enhance forecasting robustness. Additionally, experimentation was limited to selected product categories within a specific industrial context.

Future research may extend this work by integrating deep learning models such as LSTM or Transformer-based architectures, incorporating real-time data streams, and developing automated forecasting pipelines for dynamic industrial environments. Further studies may also

explore multi-product forecasting, supply chain optimization integration, and intelligent decision-support systems based on predictive analytics.

In conclusion, this research successfully established a comprehensive demand forecasting framework for the natural fibre industry. The results clearly demonstrate that hybrid forecasting models, particularly the Hybrid SARIMA approach, provide superior prediction accuracy, improved reliability, and strong industrial applicability. The study confirms that hybrid time-series and machine learning integration represents an effective and practical solution for modern demand forecasting challenges, thereby contributing both methodological advancement and industrial decision-making support.

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